

Vatsal Goel

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Founding engineer building zero-to-one cybersecurity infrastructure at an early-stage startup — solo-owning 4 production repositories spanning telemetry ingestion, anomaly detection, and zero-exfiltration pipelines in Go and Python. CNCF contributor (OpenTelemetry) and AWS Powertools maintainer. Product-minded builder with MVP-to-product-market-fit experience across system design, customer discovery, AWS/Kubernetes, and Agile delivery.

PROFESSIONAL EXPERIENCE

Founding Engineer

i4 Ops

Jul 2025 – Present

Salt Lake City, UT

- Built real-time telemetry pipeline processing 50K+ events/sec with sub-200ms P99 latency, designing a dual-mode Go agent and Python anomaly detection engine across 4 repos for zero-exfiltration enterprise security.
- Defined product roadmap and zero-trust architecture for \$15B+ market — co-designed CMMC 2.0-aligned controls, federated API gateway, and hardened VM strategy validated across defense, supply chain, and energy verticals.
- Architected event-driven pipeline using Redis Streams and PostgreSQL, replacing legacy polling agents with a unified event bus and service mesh that cut alert latency by 60%, enforced SLO/SLA targets, and eliminated dual control plane conflicts.
- Shipped human-in-the-loop export approval workflow with AI-assisted risk scoring, integrated with DLP controls and SOC 2-aligned audit logging for CMMC Level 2 compliance readiness.

Software Engineer

University of Utah

Oct 2023 – May 2025

Salt Lake City, UT

- Reduced manual data verification effort by 70% while maintaining 99.9% accuracy across 1M+ records by engineering automated Python validation workflows replacing error-prone human review.
- Improved digital preservation integrity for 1M+ archival records by building automated metadata extraction and validation pipelines in Python and Bash that caught schema violations before ingest.

Software Engineer – Full Stack

XGRO Technologies

Oct 2021 – Oct 2022

Pune, India

- Shipped 3 full-stack e-commerce platforms handling 10K+ daily transactions using React and SQL-backed REST APIs, scaling to 2x traffic growth without infrastructure changes.
- Reduced page load times by 30% and bounce rates by 12% by optimizing SQL execution plans, implementing Redis caching, and restructuring API payloads.
- Increased user engagement by 25% (session duration + CTR) by redesigning the product UI in Figma with responsive CSS layouts that improved mobile conversion.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, Java, SQL, Bash, YAML, C

Frameworks: FastAPI, Flask, Node.js, Express.js, React, REST APIs, gRPC, Pydantic, TensorFlow

Cloud & Infra: AWS, Docker, Kubernetes, Terraform, CI/CD, Git, Microservices, Service mesh, IaC, Agile

Data Systems: PostgreSQL, MongoDB, Redis, Kafka, Event-driven architecture, ETL pipelines

Security: Zero-trust architecture, DLP, SIEM, CMMC 2.0, SOC 2, NIST, IAM, Threat modeling, Observability

AI & Tooling: LLM integration, RAG pipelines, Prompt engineering, AI agents, OpenTelemetry, Vector databases

PROJECTS

AWS Powertools for Lambda (TypeScript) | TypeScript, Node.js

Jan 2025 – Present

- Implemented JSON Schema validation utilities including a `@validator` decorator; recognized as a major contributor across 3+ public releases through rigorous code review with AWS maintainers.

AWS Powertools for Lambda (Python) | Python, Pydantic

Feb 2025 – Present

- Delivered 4+ production-ready schema utilities including AppSync/WebSocket event parsing and recursive DataMasking — 100% test coverage across 40+ unit tests.

OpenTelemetry Go SDK | Go, Distributed Tracing, CNCF

Mar 2026 – Present

- Contributing to CNCF's industry-standard tracing SDK — 2 active issues addressing telemetry pipeline behavior relevant to production observability at i4 Ops.

Raft-Based Fault-Tolerant KV Store | Go, Raft, Distributed Systems

Sep 2024

- Built fault-tolerant distributed KV store using Raft consensus — leader election, log replication, and state machine execution resolving 95% of partition scenarios.

EDUCATION

University of Utah — M.S. Computer Science

May 2025

SRM Institute of Science & Technology — B.Tech of Computer Engineering

May 2023